

CIST 1401
Fall Semester 2026
Network Design Project

Your Project is to design a computer network for a law firm which is relocating to new office space in Marietta, GA. They will be located in a new large seven story building, which is just being finished. Each floor of the new building is 20,000 square feet. The building dimensions are on the attached floor plan.

- The exterior dimensions of the building are 125' x 160'.
- There is a conference room on every floor.
- The law office occupies the entire building.
- Equipment rooms are on every floor where the service provider (AT&T, Comcast, Verizon, etc.) will terminate their network equipment.
- All equipment rooms are connected together.

Things to consider

- 23 Law Partners
- 23 Administrative Assistants
- 200 Junior lawyers
- 307 support staff
- 15 IT support staff

Paper Requirements

- You are required to write a paper, using APA format, which explains your network and reasoning for making the choices which you made.
- Links for APA help are below:
 - [Sample APA Paper](#)
 - [Additional APA tutorials](#)
 - [APA Reference Generator](#)
- You are also required to develop a Power Point Presentation (15 slides max) which covers the development, cost, and your reasons for making the design choices.

Project Presentation

- ***In person class learning*** – Project will be presented to class. Final proposal (PDF and PowerPoint) must be submitted to instructor electronically.
- ***Online classes*** – Project will be submitted electronically to the instructor.

Learning Team

This is a project where you will be assigned to work with a team of your peers in class. There are deliverables related to this team assignment to help foster your growth and development working within a team dynamic.

These **DELIVERABLES** include the following:

- Group Contract - Identifies your skills and how you will work with each other.
- Group Meeting Log File - Documents your team meetings and helps you track your progress
- Individual Group Evaluation - Allows you to self-reflect on how you and your team performed together.
- Final Presentation and Paper – Your final completed project.

Each of these documents **MUST** be submitted via Blackboard Dropbox by the due date.

- Team Contract – Due 9/6/2026
- Learning Team Logs - Due 9/13/2026, 10/4/2026, & 11/1/2026
- Learning Team Evaluation - Due 11/15/2026
- Presentation and Paper - Due 11/22/2026

Design Requirements

Create a logical network map of the network. Suggested software design tools: Wire Shark, Visio, PowerPoint, Zen map (zmap), etc.... **It must not be hand drawn.**

Your design should include all

- Network Equipment – (vendors and types of equipment)
- VoIP phone services – (service provider and features included)
- Network Connectivity (wired and wireless)
- Computers (quantity, vendors and type desktop/laptop)
- Routers (quantity, vendors and type)
- Printers (quantity, vendors and type)
- Servers (and server roles) see examples of types below choose which you want to include
 - DNS, DHCP, Domain Controllers, etc.
 - Email server
 - File server
 - Web server for Internet and Intranet
- Switches (quantity, vendors and type)
- Network jacks in each office
- Cable used and types

Security is a top priority

- Physical security (badge access, security guards, etc.)
- Network security (IDS, IPS, Firewall, etc.)

Other Considerations

- Develop a data backup solution (cloud storage)
- Maintenance services and support (ongoing, 24-month, 36-month, etc. specify the duration and cost)
- Specify the cost of your project, with breakdown cost of items from 1 through 9 above. Your budget cap is \$4.25 million. Prioritize purchases, as you may be asked to scale back due to lack of funding.

Project Presentation

Final Project is due in week 15. The following items must be submitted via Blackboard Drop Box.

- Each Team member Must individually submit the group assignments in order to receive their individual grades.
- This will consist of the formal paper which is an APA formatted Word document.
- PowerPoint Presentation with a maximum of 15 slides
- The Learning Team Evaluation

